ANTHONY MICHEAL IDEDIA

Data Science & AI | Monitoring, Evaluation and Data Insights | Health and Development Programmes | Nigeria

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PROFESSIONAL SUMMARY

Data science and analytics professional with 6 years building the pipelines, models and evidence behind donor-funded health, livelihoods, climate and inclusive-finance programmes across Nigeria and Africa. Currently handles the research and data workstream at Innovision Global Consulting, an international advisory and management consulting firm operating in 22 countries, owning the full analytical cycle: data engineering, statistical and machine learning analysis, dashboards, data quality and governance standards, and the translation of technical findings into decisions for government, donor, investor and private-sector audiences. Builds multi-source pipelines in Python and Linux, has processed 1.2 million high-resolution data points through automated QA/QC workflows, and fine-tuned a deep learning segmentation model to a 10 per cent gain in accuracy over baseline. Designs AI-assisted production workflows, including a scoring architecture that runs two large language models against a deterministic scorer and reconciles them through a consensus layer, so no model output reaches a user unvalidated. Co-author of a national malaria policy report whose pricing recommendations were taken up by NMEP stakeholders, and author of a public finance evidence series built on a reproducible notebook that re-derives every published figure from its original source file. MSc Bioinformatics (Distinction), with a published research record.

AREAS OF EXPERTISE

- **Data Science and Statistical Modelling:** classification and regression modelling, price elasticity estimation, longitudinal and multi-wave trend analysis, cross-sectional inference with multiple-comparison correction, and deep learning model fine-tuning.
- **Data Engineering and Pipelines:** Python and Linux pipeline design, multi-source integration across APIs, RSS, web sources, form platforms and cloud spreadsheets, automated extraction from large PDF and Excel corpora, entity resolution and de-duplication.
- **AI-Enabled Workflows and Responsible AI:** large language models embedded in production workflows, multi-engine consensus validation of AI-assisted output, structured output design, model lifecycle and API cost management.
- **Dashboards and Decision Support:** Power BI dashboards for KPI tracking and management reporting, reproducible charting in Python, and native editable Office charts for client-facing reports.
- **Data Quality, Validation and Governance:** verification protocols, provenance ledgers, assertion-based validation against published anchors, metric taxonomies, confidence flagging, and documentation standards applied across partner and third-party datasets.
- **Monitoring, Evaluation and Research:** study scoping and research design, mixed-methods instruments, baseline, midline and endline analysis, evidence synthesis, and policy analysis.
- **Digital Data Collection and Health Systems Data:** ODK, SurveyCTO, KoboCollect, Typeform and CAPI, including form scripting, skip logic, constraint and validation rules; national household, outlet and facility surveys under PEPFAR, WHO and national ethical protocols.
- **Communication and Capability Building:** technical findings for non-technical and senior stakeholders, policy briefs and knowledge products, mentoring and structured capacity-building programmes, and technical input into proposals, concept notes and partnerships.

PROFESSIONAL EXPERIENCE

Research and Data Analyst | Innovision Global Consulting International (Nigeria) Ltd. *Nov 2025 to Present*

- Performed the research and data workstream across the Nigeria portfolio, owning study scoping, pipeline design, analysis and client-facing deliverables alongside portfolio managers, practice area leads and the Regional Director for Africa.
- Design and run multi-source pipelines in Python (pandas, openpyxl, pdfplumber, pdftotext) that parse several hundred annual reports, disclosures and financial workbooks into structured, analysis-ready datasets, with entity resolution and de-duplication across roughly 983 raw records drawn from 35 source sheets.
- Set the data quality, validation and documentation standards applied to partner and third-party datasets: a metric taxonomy separating assets under management, committed and deployed capital; a confidence flag on every record; page-level source citation; and a rule against inferring, converting or rounding published figures.
- Build automated data collection and integration workflows in n8n Cloud, connecting form platforms, email, APIs and a live multi-tab data store, with parsing, completeness scoring and conditional routing running unattended across 158 target organisations.

- Run the statistical work across the portfolio: logistic regression and price elasticity estimation on national health-market data, multi-wave harmonisation of 15 or more datasets covering 1,800 respondents from 2018 to 2024, and 36-state cross-sectional analysis with multiple-comparison correction applied before any relationship is reported.
- Present technical findings to non-technical, senior and client audiences through policy reports, dissemination materials and pitch decks, and provide technical input into proposals and partnerships, including the service proposition, pitch deck and ranked 90-organisation prospect pipeline for the firm's impact evaluation retainer offer.
- Document methods so colleagues can run them: reproducible analysis notebooks, written build guides for the automation workflows, and placement guides used by the team to produce client deliverables themselves.

Survey Enumerator and Field Laboratorian | Nigeria ART Impact Survey, NASCP / NCDC / PEPFAR *2024*

- Conducted structured household interviews and biospecimen collection using CAPI software under national ethical protocols, PEPFAR data quality standards and WHO-aligned chain-of-custody documentation.
- Worked within multi-agency field teams (NASCP, NCDC, NACA, Lagos State Ministry of Health) to keep survey implementation consistent across facility and community levels for a national endline evaluation.

Graduate Research Assistant | Covenant University, World Bank CApIC-ACE Project *Oct 2023 to Aug 2025*

- Managed and processed 1.2 million high-resolution data points through Linux-based pipelines, applying QA/QC protocols at each stage to keep datasets clean and analysis-ready across the project cycle.
- Wrote Python and Bash automation that cut data processing time by 18 per cent, helping the team meet World Bank reporting deadlines across multiple cycles.
- Fine-tuned an AI segmentation model (MobileSAM) to a 10 per cent gain in accuracy over baseline and presented the technical findings to cross-functional and non-technical stakeholders.

Data Training and Development Coordinator | AlphaTrack Intelligence *Dec 2023 to Oct 2025*

- Served as internal research and evaluation lead for the learning function, running organisational needs assessments and designing structured, evidence-based capability programmes including mentoring and coaching.
- Built Power BI dashboards tracking training KPIs (completion, skill scores, feedback) and produced quarterly evaluation reports; programme efficacy improved 21 per cent and team productivity 36 per cent.

Data Analyst, Research Projects | AlphaTrack Intelligence *Oct 2022 to Nov 2023*

- Delivered quantitative and qualitative research analysis in Python, SPSS and GraphPad, improving the predictive accuracy of research findings by 27 per cent, and produced evidence-based reports for executive decision-making.

Independent Field Monitor | World Health Organization *Jun to Oct 2022*

- Supported a national vaccination campaign in Ogun State: community mobilisation, health worker training, field data monitoring and structured reporting to WHO supervisors.

SELECTED PROGRAMME EXPERIENCE

Impact Investors Foundation: Nigeria Impact Investment Market Sizing *Nigeria, 2025 to 2026; approximately 62 investor organisations; approximately 332 primary sources*

Objective: Produce the first comprehensive quantification of impact-focused capital flows into Nigeria, 2020 to 2026, to a standard that survives public and investor scrutiny.

- Analyst. Designed the research protocol and the metric taxonomy separating assets under management, committed capital and deployed capital, so every figure remains comparable and traceable to source.
- Built the extraction pipeline across development finance institution reports, fund disclosures and portfolio databases, consolidating outputs into a multi-sheet master workbook covering annual flows, stock, sector allocation and deal-level records.
- Diagnosed and corrected structural faults in the inherited dataset, including double-counted syndicate deals, metric-class confusion across owned, guaranteed and arranged capital, reference-date drift, and a currency conversion approach that would have distorted domestic institution figures.
- Designed and built the primary data collection system: a 42-screen instrument with conditional logic feeding an n8n workflow that sequences outreach, parses responses, scores completeness into tiers and routes follow-up automatically across a 158-organisation contact base.

Malaria Consortium and National Malaria Elimination Programme: Nationwide Market Survey on Malaria Case Management and Prevention *Nigeria, 2018 to 2025 series; all six geopolitical zones*

Objective: Measure demand, supply, pricing and access dynamics in Nigeria's malaria commodity market to inform national distribution and pricing policy.

- Led the analysis of a national household and outlet survey covering 6,000 records: SPSS cleaning and validation, data quality checks, price elasticity modelling and logistic regression.

- Built a multi-wave analysis framework in Python (pandas, statsmodels) and SPSS, harmonising 15 or more datasets covering 1,800 respondents from 2018 to 2024; the resulting trend analysis informed national malaria commodity distribution decisions.
- Co-authored the policy report whose micro-pricing recommendations were taken up by NMEP stakeholders, and contributed to the national dissemination materials.

Danish Refugee Council and European Union: Green Labour Market and Climate-Smart Agriculture Baseline *Sokoto State, Nigeria, 2025; 442 households across four LGAs*

Objective: Establish a baseline on livelihoods, food security, green skills demand and climate-smart agriculture adoption to guide EU-funded economic recovery programming.

- Scripted the ODK instruments and ran the full analysis covering livelihoods, assets, agriculture and yields, a nine-item food insecurity module, green skills demand and vocational training access, disaggregated by displacement status, gender of household head, disability and income quintile.
- Built the classification logic behind the report's central claims, including climate-smart agriculture adoption status, a resilience category coded from a 111-strategy library of active and passive coping responses, and a training misalignment measure cross-referencing sector trained against current livelihood.
- Adopters of climate-smart practices returned roughly double the yield per hectare of non-adopters, 883 kg against 433 kg, the report's headline agricultural finding.

Innovision: Nigeria Evidence Series on Public Finance and the Cost of Living *Nigeria, 2022 to 2026; 36 states and the FCT*

Objective: Test whether Nigeria's reform-era revenue windfall reached households, using only primary published data, and demonstrate analytical capability to public-sector and donor audiences.

- Authored the third report in the series and built the reproducible notebook behind it: every figure is re-derived at run time from original NBS, CBN, OAGF, NUPRC, DMO and NEITI files, validated with assertions against independently published anchors, and logged in a provenance ledger giving each figure its basis, source file and derivation.
- Computed the 21.90 trillion naira all-tier federation total directly from twelve monthly OAGF disbursement workbooks rather than carrying a cited constant, and spliced a rebased consumer price index series across the 2024 rebasing break.
- Ran the 36-state cross-sectional analysis relating allocation per person to the price shock households face, and after a methodological audit applied Bonferroni correction across five cross-sections, reporting the affected result as not surviving correction rather than retaining it.
- Delivered the report with ten native editable Office charts, each backed by an embedded worksheet, so readers can interrogate the underlying figures directly.

Covenant University and World Bank CApIC-ACE: AI Model Development and Genomic Data Pipelines *Nigeria, 2023 to 2025; 1.2 million high-resolution data points*

Objective: Build reproducible high-throughput analysis capacity and apply deep learning to histopathology image segmentation.

- Fine-tuned MobileSAM for few-shot segmentation of prostate cancer in histopathology images, achieving a 10 per cent gain in accuracy over baseline; awarded a Distinction.
- Built Linux-based processing pipelines with QA/QC protocols at each stage, with Python and Bash automation that cut processing time by 18 per cent.
- Co-authored genomics research on the architecture of three major cancers in African-ancestry populations and a Bayesian meta-analysis of SRD5A2 polymorphisms and prostate cancer risk.

Innovision: Opportunity Radar, an automated multi-source intelligence pipeline *Nigeria and Africa portfolio, 2026; 70 or more monitored sources*

Objective: Replace manual scanning of donor, multilateral and government procurement portals with an automated, scored and routed intelligence feed.

- Classified 184 source URLs across 111 domains into an accessibility registry by method (API, RSS, scrape, email alert), and ranked the highest-yield sources against historical opportunity data.
- Built the pipeline in n8n Cloud: scheduled ingestion, normalisation, de-duplication against a seen-item store, keyword filtering, scoring, consensus reconciliation and routed weekly digests to four portfolio leads.
- Designed a triple-engine scoring architecture running two large language models and a deterministic keyword scorer as parallel branches, merged and reconciled by a consensus node, so no single model decides what reaches a colleague. Managed API cost, error handling and model lifecycle, including migration off a deprecated model.

KEY SKILLS

- **Programming and analysis:** Python (pandas, NumPy, statsmodels, matplotlib), SPSS, GraphPad, advanced Excel including pivot analysis, Linux command line and Bash scripting.
- **Modelling:** logistic regression and classification, price elasticity and regression estimation, correlation and cross-sectional analysis, multiple-comparison correction, longitudinal and multi-wave harmonisation, deep learning model fine-tuning for image segmentation.
- **Data engineering:** extraction and transformation from PDF, Excel and API sources (pdfplumber, pdftotext, openpyxl), entity resolution and token-based fuzzy matching, de-duplication, and workflow automation in n8n Cloud.
- **AI and LLM tooling:** Anthropic and Google model APIs in production workflows, multi-engine consensus validation, structured output design, prompt and rule design, model deprecation and cost management.
- **Visualisation and reporting:** Power BI dashboards, Python charting, native editable Office charts injected at OOXML level, policy briefs, technical reports and pitch materials.
- **Data collection:** ODK, SurveyCTO, KoboCollect, Typeform, CAPI; form scripting, conditional logic, constraint and validation rules, GPS validation.
- **Qualitative analysis:** ATLAS.ti; thematic coding, inter-coder consistency, and synthesis of key informant and focus group data with survey findings.
- **Governance and standards:** reproducible analysis notebooks with assertion-based validation, provenance ledgers, PEPFAR data quality and WHO-aligned chain-of-custody protocols, national ethical approval processes, OECD-DAC evaluation criteria.

EDUCATION

- MSc Bioinformatics (Distinction), Covenant University, Ota; World Bank-affiliated CApIC Centre of Excellence | 2025
- BMLS Chemical Pathology, Ambrose Alli University, Ekpoma, Edo State | 2019

PROFESSIONAL CERTIFICATION

- Data Analytics and Business Intelligence, Dataleum, Lagos | 2022

PUBLICATIONS

- David, E., Anthony, M. I., et al. (2025). Deciphering the Genomic Architecture of Three Major Cancers in African-Ancestry Populations. doi:10.64898/2025.12.19.25342629
- Isewon, I., Anthony, M., et al. (2025). Bayesian Meta-Analysis of SRD5A2 Polymorphisms and Prostate Cancer Risk in African Populations. NIPES Journal of Science and Technology Research, 7(1). doi:10.37933/nipes/7.4.2025.si278
- Anthony, M. I. (2025). Adapting MobileSAM for Few-Shot Segmentation of Prostate Cancer in Histopathology Images. MSc Dissertation, Covenant University.
- Lead author, systematic review on the prevalence of teenage pregnancy in Nigeria. Completed, in preparation for submission.

ADDITIONAL INFORMATION

- **Languages:** English (full professional proficiency).
- **Systems:** Linux command line, reproducible analysis notebooks, MS Office and Google Workspace.
- **Research ethics and data handling:** work delivered under national ethical approval, PEPFAR data quality standards and WHO-aligned chain-of-custody documentation, with consent and confidentiality protocols applied throughout.
- **Availability:** based in Lagos and open to Abuja relocation and hybrid working.

References available on request.